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➢Title

Understanding Customer Preferences and Restaurant Trends: An In-Depth Analysis Using the Zomato Dataset

➢Executive Summary

This report provides a comprehensive analysis of customer preferences and restaurant trends within the **Zomato dataset**. The objective of this analysis is to explore and understand the different factors influencing customer satisfaction and popularity of restaurants. Key variables include restaurant location, cuisine types, price range, and customer ratings. This study highlights trends and patterns that could be valuable for restaurant owners, industry stakeholders, and marketers aiming to align business strategies with customer expectations and market demands.

➢Dataset Overview

Link : https://www.kaggle.com/code/payamamanat/zomato-restaurant-analysis

The Zomato dataset consists of a range of features capturing various aspects of restaurants, including:

Field	Description
Restaurant ID	Unique identifier for each restaurant
Name	Name of the restaurant
Location	The geographic area or city where the restaurant is located
Cuisine	Types of cuisine served at each restaurant
Average Cost	Average dining cost for two people
Rating	Customer ratings on a scale from 1 to 5, indicating satisfaction
Reviews Count	The number of reviews each restaurant has received, often reflecting popularity

Each feature plays an important role in understanding customer choices and restaurant performance across regions. The analysis explores how these factors impact customer satisfaction and offers insights into potential areas of improvement.

➢Problem Statements

1. Identification of the Most Common Types of Restaurants on Zomato

Objective: To ascertain the most prevalent types of restaurants listed on Zomato, thereby highlighting customer preferences and the popularity of various dining experiences.

Expected Outcome: Recognize the top 10 restaurant types by count to determine the primary focus areas within the restaurant market.

2. Evaluation of the Most Expensive Restaurants on Zomato.

Objective: To identify the top 10 restaurants with the highest average cost for two people, providing insight into establishments catering to premium market segments.

Expected Outcome: List the top 10 most expensive restaurants to understand pricing trends and the positioning of premium dining establishments.

3. Analysis of the Distribution of Customer Ratings Across Restaurants

Objective: To examine the distribution of the number of ratings received by restaurants, shedding light on customer engagement and the popularity of various establishments.

Expected Outcome: Identify the frequency distribution of ratings to understand overall customer interaction levels with restaurants on the platform.

4. Identification of the Most Popular Cuisines on Zomato

Objective: To identify the top 10 most frequently offered cuisines, providing insights into the culinary preferences of customers and the demand for specific cuisine types.

Expected Outcome: Highlight the top cuisines to understand customer preferences and guide restaurants in selecting menu offerings.

5. Identification of Budget-Friendly Restaurants on Zomato

Objective: To list the 10 least expensive restaurants based on the average cost for two people, providing insights into establishments catering to cost-conscious customers.

Expected Outcome: Identify budget-friendly restaurants, offering insights for attracting customers who prioritize affordability.

6. Analysis of Online Ordering Availability Among Restaurants

Objective: To assess the prevalence of online ordering availability, highlighting the importance of digital engagement in today's market.

Expected Outcome: Understand the proportion of restaurants offering online ordering to meet digital customer expectations.

7. Analysis of Table Booking Availability Across Restaurants

Objective: To evaluate the availability of table booking, particularly for formal dining settings where customers may expect reserved seating.

Expected Outcome: Determine the distribution of table booking availability to understand customer expectations in formal dining.

8. Analysis of Areas with the Highest Concentration of Restaurants

Objective: To identify the top 10 areas with the highest number of restaurants, providing insights into market saturation and customer demand by location.

Expected Outcome: Recognize high-density areas with restaurant competition or high demand.

9. Examination of the Relationship Between Ratings and Average Cost

Objective: To assess whether higher pricing correlates with better ratings, indicating customer perceptions of value.

Expected Outcome: Determine if there is a positive correlation between cost and ratings, helping businesses understand the impact of pricing on customer satisfaction.

➢ Methodology

1. Data Preprocessing
2. Descriptive Statistics
3. Data Visualization
4. Correlation Analysis

➢Code and Analysis

1. Data Preprocessing and Exploration

Data preprocessing is essential for ensuring the dataset is clean, accurate, and ready for analysis. It involves handling missing values, removing duplicates, and examining the dataset's general structure. These steps ensure our analysis is based on reliable data.

A. Loading Necessary Libraries

```
#Loading Necessary Libraries
library(tidyr)     # For data tidying
library(dplyr)     # For data manipulation
library(ggplot2)   # For data visualization
```

a) tidyr: For data tidying, allowing us to reshape and organize data more effectively.

b) dplyr: For data manipulation, enabling filtering, summarization, and transformations.

c) ggplot2: For data visualization, to create plots and graphs that visually represent trends and distributions.

Purpose: This code loads three essential R libraries to handle, manipulate, and visualize the Zomato dataset.

B. Loading the Data

```
#Loading the Data
df = read.csv("C:/Downloads/zomato.csv")
```

Purpose: This line imports the Zomato dataset into R from the specified file path and stores it in a data frame named `df`.

C. Checking Column Names

```
#Checking Column Names
colnames(df)
```

Purpose: This line lists all column names in the dataset, providing an initial view of the data structure.

Output:

```
[1] "X"                    "Unnamed..0"           "restaurant.name"       "restaurant.type"
[5] "rate..out.of.5."      "num.of.ratings"       "avg.cost..two.people." "online_order"
[9] "table.booking"        "cuisines.type"        "area"                  "local.address"
```

Explanation:

This list represents all the columns present in the Zomato dataset.

Each column name gives insight into the type of information it holds:

- **"X" and "Unnamed..0"**: These columns appear to be generated during data export or storage and may not contain valuable information for analysis.
- **"restaurant.name"**: The name of each restaurant, helping in restaurant-specific analyses.
- **"restaurant.type"**: The type of restaurant (e.g., dine-in, delivery, or takeaway).
- **"rate..out.of.5."**: Customer rating out of 5, representing customer satisfaction levels.
- **"num.of.ratings"**: The total number of ratings received, indicating the restaurant's popularity.
- **"avg.cost..two.people."**: Average cost for two people, helping analyze price ranges across restaurants.
- **"online_order"** and **"table.booking"**: Indicate whether online ordering and table booking options are available, which may impact customer preferences.
- **"cuisines.type"**: The type of cuisines offered, useful for understanding cuisine popularity.
- **"area"** and **"local.address"**: Geographic information, allowing for location-based analysis.

D. Dropping Unnecessary Columns

```
#Dropping Unnecessary Columns
columns_to_drop = c("Unnamed..0.1", "Unnamed..0")
existing_columns = columns_to_drop[columns_to_drop %in% colnames(df)]
df = df %>% select(-all_of(existing_columns))
```

- **Purpose**: This code removes any unnecessary or placeholder columns, ensuring the dataset is clean and concise.

E. Quick Data Scan

```
#Quick Data Scan
glimpse(df)
summary(df)
```

- **Purpose**: These lines provide an overview of the dataset's structure and summary statistics.

```
> glimpse(df)
Rows: 7,105
Columns: 11
$ X                    <int> 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 1…
$ restaurant.name      <chr> "#FeelTheROLL", "#L-81 Cafe", "#refuel", "'@ Biryani Central", …
$ restaurant.type      <chr> "Quick Bites", "Quick Bites", "Cafe", "Casual Dining", "Casual …
$ rate..out.of.5.      <dbl> 3.4, 3.9, 3.7, 2.7, 2.8, 3.4, 4.1, 2.8, 3.2, 3.7, 3.5, 4.4, 4.1…
$ num.of.ratings       <int> 7, 48, 37, 135, 40, 37, 305, 40, 49, 41, 22, 2861, 353, 119, 17…
$ avg.cost..two.people. <dbl> 200, 400, 400, 550, 700, 200, 700, 300, 300, 450, 300, 1500, 20…
$ online_order         <chr> "No", "Yes", "Yes", "Yes", "Yes", "No", "Yes", "No", "Yes", "No…
$ table.booking        <chr> "No", "No", "No", "No", "No", "No", "No", "No", "No", "No", "No…
$ cuisines.type        <chr> "Fast Food", "Fast Food, Beverages", "Cafe, Beverages", "Biryan…
$ area                 <chr> "Bellandur", "Byresandra,Tavarekere,Madiwala", "Bannerghatta Ro…
$ local.address        <chr> "Bellandur", "HSR", "Bannerghatta Road", "Marathahalli", "Bella…
```

The `glimpse(df)` function provides a detailed view of the dataset's structure, revealing important information such as the number of rows, columns, data types, and sample data for each column.

Output Summary:

- **Rows**: The dataset contains **7,105 rows**, representing individual restaurant entries.
- **Columns**: There are **11 columns** in total, each representing different attributes of the restaurants.

Significance:

- This structured overview provides critical insights into each column's purpose, data type, and sample values, helping us plan the next steps for data cleaning and analysis.
- Columns such as `rate..out.of.5.`, `num.of.ratings`, and `avg.cost..two.people.` are particularly useful for understanding customer satisfaction, engagement, and price points.
- The geographic columns (`area` and `local.address`) allow for location-based analysis, while `restaurant.type` and `cuisines.type` help in categorizing restaurants by style and cuisine.

```
> summary(df)
       X        restaurant.name    restaurant.type    rate..out.of.5. num.of.ratings
 Min.   :   0   Length:7105        Length:7105        Min.   :1.800   Min.   :    1.0
 1st Qu.:1776   Class :character   Class :character   1st Qu.:3.200   1st Qu.:   16.0
 Median :3552   Mode  :character   Mode  :character   Median :3.500   Median :   40.0
 Mean   :3552                                         Mean   :3.514   Mean   :  188.9
 3rd Qu.:5328                                         3rd Qu.:3.800   3rd Qu.:  128.0
 Max.   :7104                                         Max.   :4.900   Max.   :16345.0
                                                      NA's   :68
 avg.cost..two.people. online_order       table.booking      cuisines.type
 Min.   :  40.0        Length:7105        Length:7105        Length:7105
 1st Qu.: 300.0        Class :character   Class :character   Class :character
 Median : 400.0        Mode  :character   Mode  :character   Mode  :character
 Mean   : 540.3
 3rd Qu.: 600.0
 Max.   :6000.0
 NA's   :57
     area           local.address
 Length:7105        Length:7105
 Class :character   Class :character
 Mode  :character   Mode  :character
```

The `summary(df)` function provides an overview of essential statistics for each column in the dataset, helping us understand the data distribution, detect potential outliers, and identify columns with missing values.

Key Takeaways:

- **Missing Values**: Missing values are present in `rate..out.of.5.` (68 entries) and `avg.cost..two.people.` (57 entries). These values will need to be imputed or handled to ensure analysis accuracy.
- **Distribution Insights**: Most ratings are centered around the 3.5 mark, and the average cost per two people tends to fall between 300 and 600. This indicates that the majority of restaurants are mid-rated and mid-priced.
- **Outliers**: The maximum values for `num.of.ratings` and `avg.cost..two.people.` suggest some entries are outliers, possibly representing highly popular or premium-priced restaurants.

F. Handling Missing Values

```
#Handling Missing Values
missing_values <- df %>% summarise(across(everything(), ~sum(is.na(.))))
print("Missing Values per Column:")
print(missing_values)
```

- **Purpose**: This code identifies the number of missing values in each column, enabling data quality assessment and highlighting where imputation is needed.
  - **`summarise(across(everything(), ~sum(is.na(.))))`**: Counts the missing values in each column of `df` by applying `sum(is.na(.))` across all columns.
  - Printing `missing_values` provides a summary, showing how many values are missing in each column.

```
> print("Missing Values per Column:")
[1] "Missing Values per Column:"
> print(missing_values)
  X restaurant.name restaurant.type rate..out.of.5. num.of.ratings avg.cost..two.people.
1 0               0               0              68              0                    57
  online_order table.booking cuisines.type area local.address
1            0             0             0    0             0
```

Key Columns with Missing Values: `rate..out.of.5.` and `avg.cost..two.people.`. These variables are critical for analyzing customer satisfaction and pricing and will require imputation or other handling methods to ensure data accuracy.

```
df <- df %>%
  mutate(across(c(`rate..out.of.5.`, `avg.cost..two.people.`), ~replace_na(., mean(., na.rm = TRUE))))
```

- **Purpose**: This step imputes missing values in the columns `rate..out.of.5.` and `avg.cost..two.people.` by replacing them with the mean of each column.

  - **`mutate(across(...))`**: Applies transformations to specific columns.
  - **`replace_na(., mean(., na.rm = TRUE))`**: Uses the column's mean (ignoring NA values) to replace missing data. Imputing with the mean ensures data consistency while minimizing bias in statistical calculations.
  -

G. Checking for Duplicates

```
#Checking for Duplicates:
duplicate_count = sum(duplicated(df))
print(paste("Number of duplicate rows:", duplicate_count))
```

- **Purpose**: This code identifies duplicate rows, helping to prevent data redundancy and inaccuracies in analysis.

  - **`sum(duplicated(df))`**: Counts the number of duplicated rows in `df`, giving a total count of duplicated records.
  - Printing the result informs you if duplicates exist, and if necessary, further steps (like removing duplicates) can be taken.

```
> print(paste("Number of duplicate rows:", duplicate_count))
[1] "Number of duplicate rows: 0"
```

No duplicate rows found.

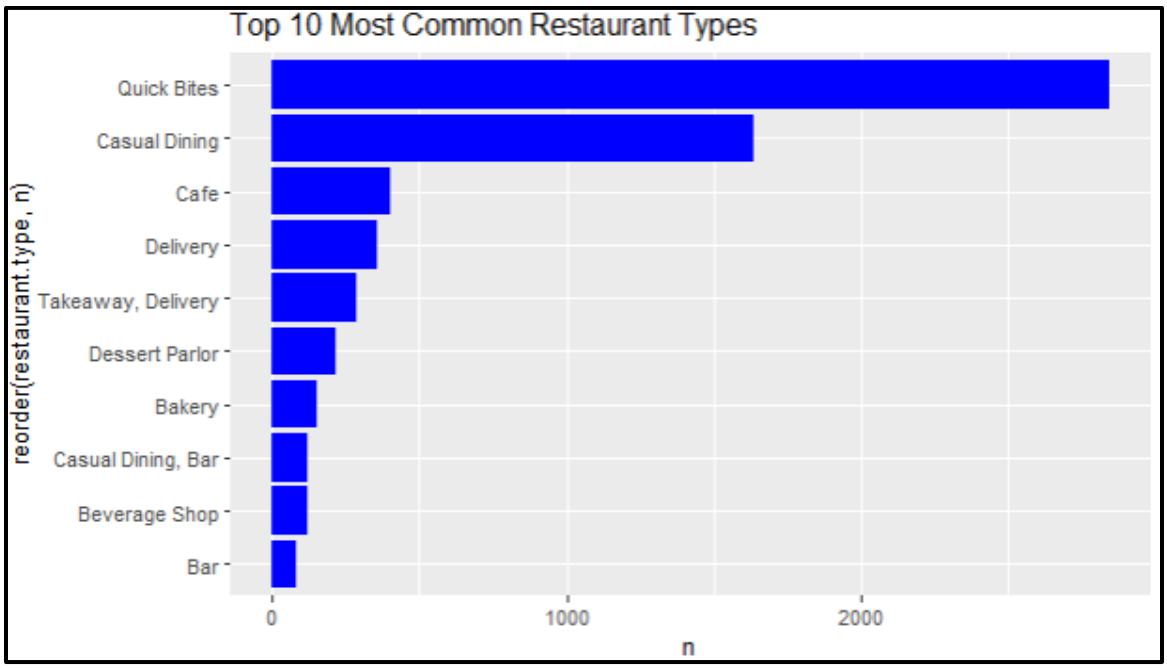
2. Identification of the Most Common Types of Restaurants on Zomato

Understanding the most common types of restaurants provides valuable insights into customer behavior and preferences. By examining the top restaurant types, such as "Quick Bites" and "Casual Dining," this analysis helps pinpoint the categories that draw the largest customer base. This insight is crucial for new entrants deciding on restaurant formats and for existing businesses seeking to expand.

R Code:

```
ggplot(df %>% count(restaurant.type, sort = TRUE) %>% head(10),
       aes(x = reorder(restaurant.type, n), y = n)) +
  geom_bar(stat = "identity",fill="Blue") + coord_flip() +
  labs(title = "Top 10 Most Common Restaurant Types")
```

Output:



3. Evaluation of the Most Expensive Restaurants on Zomato

Analyzing the top 10 most expensive restaurants on Zomato allows us to understand which establishments target a more affluent, high-income customer base. By identifying these high-cost restaurants, this analysis provides insights into the market's premium segment, where customer expectations for quality and exclusivity are higher. For new upscale restaurants, entering this segment requires a focus on unique dining experiences and superior service to meet premium customer standards.

R Code:

```
ggplot(df %>% arrange(desc(avg.cost..two.people.)) %>% head(10),
       aes(x = reorder(restaurant.name, avg.cost..two.people.), y = avg.cost..two.people.)) +
  geom_bar(stat = "identity",fill="red") + coord_flip() +
  labs(title = "Top 10 Most Expensive Restaurants")
```

Output:

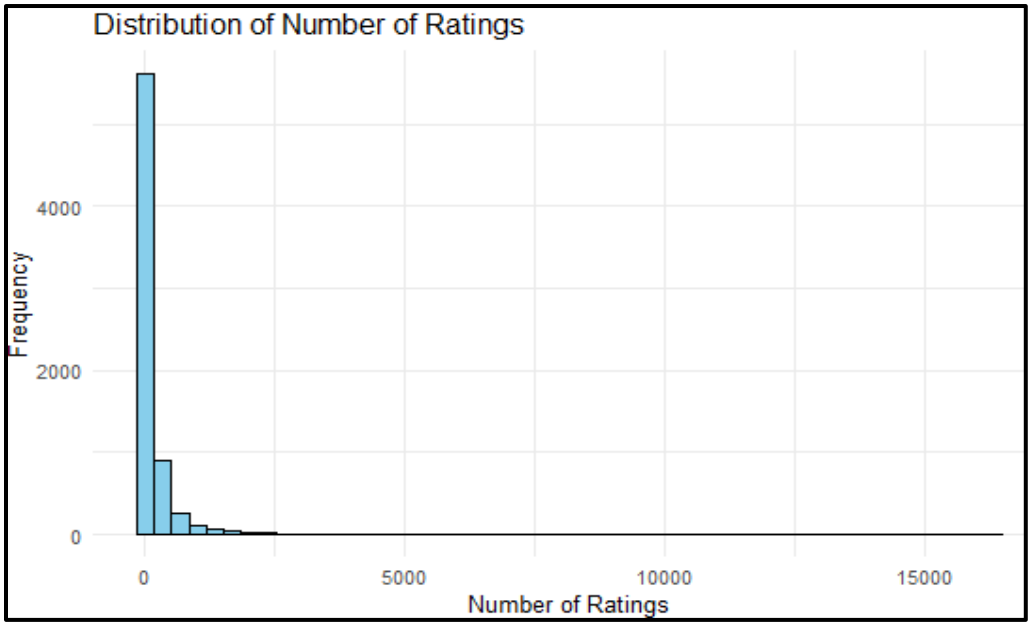
4. Analysis of the Distribution of Customer Ratings Across Restaurants

The distribution of the number of ratings reveals the level of customer engagement with restaurants. Establishments with high engagement levels are likely to have a loyal customer base and a strong reputation on Zomato. By identifying patterns in customer ratings, this analysis provides a snapshot of customer loyalty and interaction levels, guiding strategies for building brand presence and attracting repeat customers.

R Code:

```
ggplot(df, aes(x = num.of.ratings)) +
  geom_histogram(bins = 50, fill = "skyblue", color = "black") +
  labs(title = "Distribution of Number of Ratings", x = "Number of Ratings", y = "Frequency") +
  theme_minimal()
```

Output:



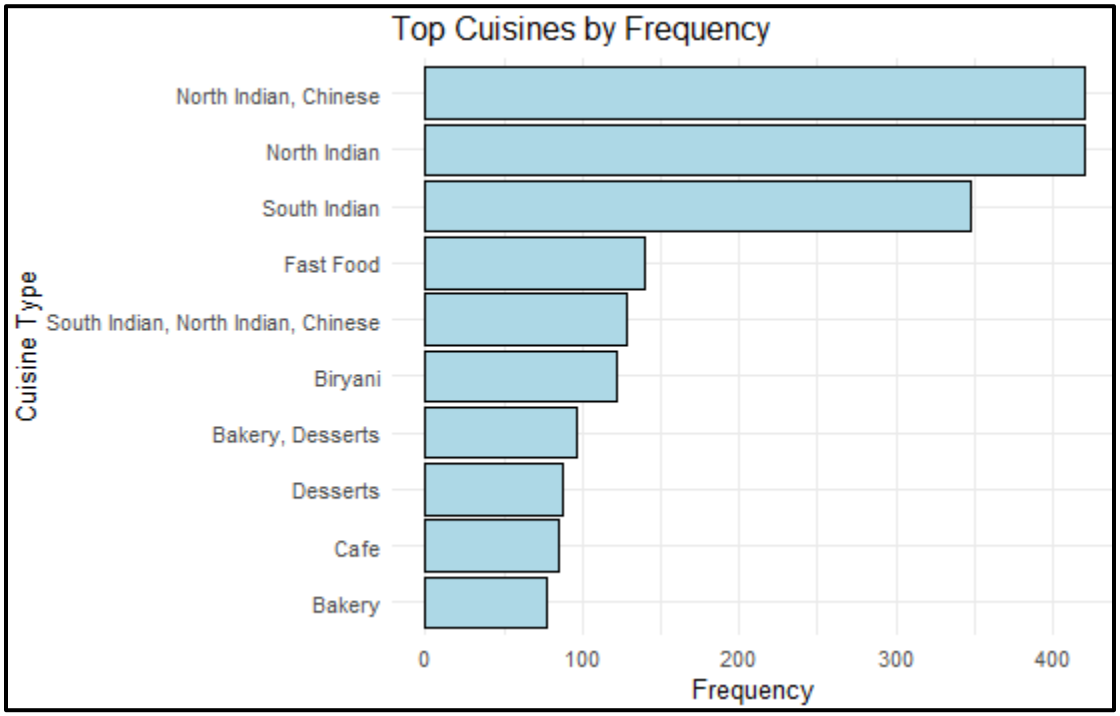
5. Identification of the Most Popular Cuisines on Zomato

By analyzing the top 10 most popular cuisines, this visualization uncovers which culinary options are most in demand. Knowing the most frequently offered cuisines can help restaurants align their menus with customer preferences, thereby attracting a wider audience. For instance, a new restaurant could consider including popular cuisine types to increase its appeal.

R Code:

```
# Count the frequency of each cuisine type and select the top 10
top_cuisines <- df %>%
  count(cuisines.type, sort = TRUE) %>%
  top_n(10, n)

# Create a bar plot of the top 10 cuisines by frequency
ggplot(top_cuisines, aes(x = reorder(cuisines.type, n), y = n)) +
  geom_bar(stat = "identity", fill = "lightblue", color = "black") +
  coord_flip() +
  labs(title = "Top Cuisines by Frequency", x = "Cuisine Type", y = "Frequency") +
  theme_minimal()
```

Output:



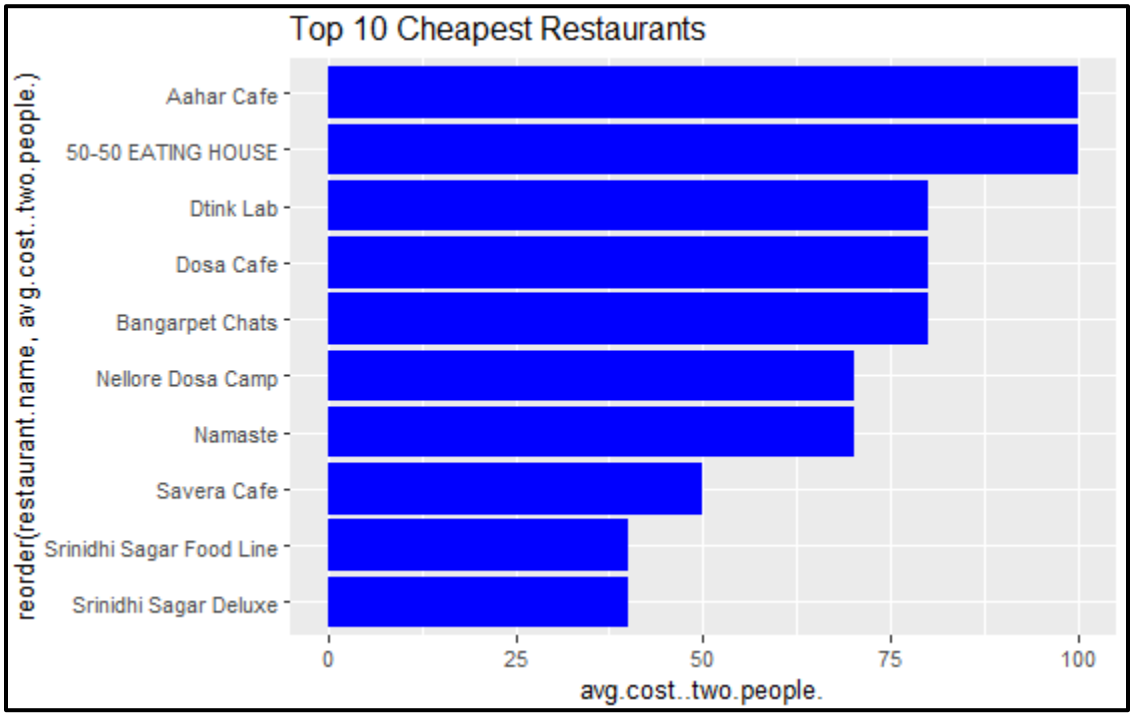
6. Identification of Budget-Friendly Restaurants on Zomato

Understanding which restaurants are the least expensive provides insights into the budget-friendly segment of the restaurant market. Restaurants that cater to cost-sensitive customers can leverage this information to emphasize value and affordability in their promotions. This analysis supports restaurants in tailoring offerings to appeal to a broader customer demographic seeking affordable dining options.

R Code:

```
ggplot(df %>% arrange(avg.cost..two.people.) %>% head(10),
       aes(x = reorder(restaurant.name, avg.cost..two.people.), y = avg.cost..two.people.)) +
  geom_bar(stat = "identity",fill="blue") + coord_flip() +
  labs(title = "Top 10 Cheapest Restaurants")
```

Output:



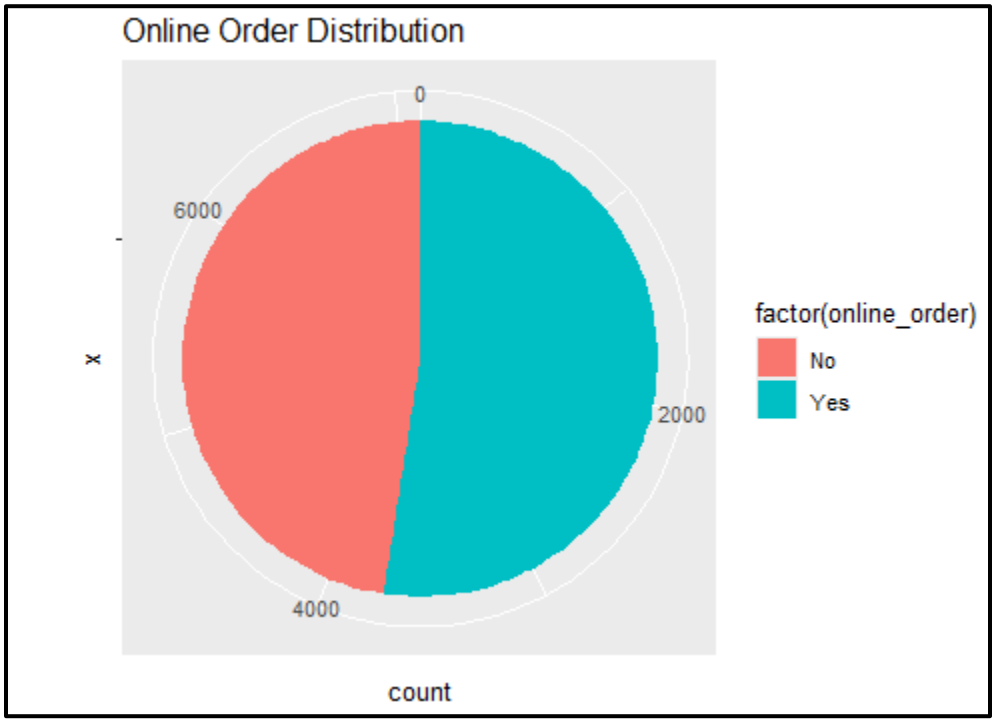
7. Analysis of Online Ordering Availability Among Restaurants

Online ordering has become a key feature in today's restaurant industry, driven by customer demand for convenience. By analyzing the distribution of online ordering availability, this visualization offers insights into how widely adopted this service is among restaurants. Restaurants not offering online ordering may be missing opportunities to attract a larger, tech-savvy customer base.

R Code:

```
ggplot(df, aes(x = "", fill = factor(online_order))) +
  geom_bar(width = 1) + coord_polar("y") +
  labs(title = "Online Order Distribution")
```

Output:



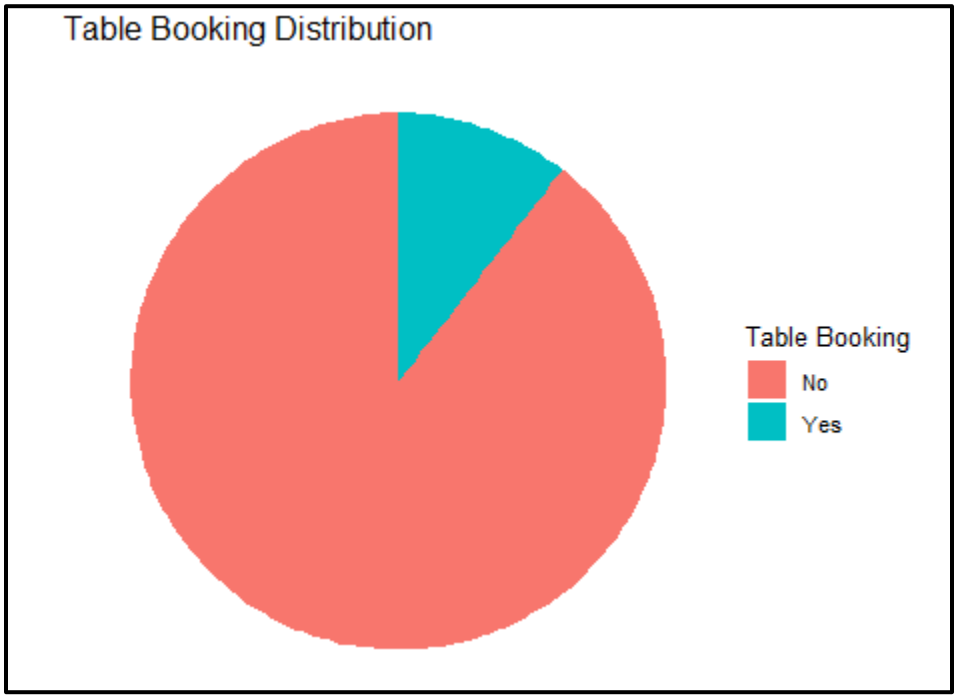
8. Analysis of Table Booking Availability Across Restaurants

The option to reserve a table can be essential for formal dining establishments, where customers may expect a guaranteed seating arrangement. This analysis identifies the distribution of table booking availability across various restaurant types, highlighting its importance for customer satisfaction in specific segments. Establishments in the formal dining space may benefit from offering reservations to meet these customer expectations.

R Code:

```
# Count the distribution of table booking options
table_booking_dist <- df %>%
  count(table.booking)

# Create a pie chart
ggplot(table_booking_dist, aes(x = "", y = n, fill = factor(table.booking))) +
  geom_bar(width = 1, stat = "identity") +
  coord_polar("y") +
  labs(title = "Table Booking Distribution", fill = "Table Booking") +
  theme_void()
```

Output:



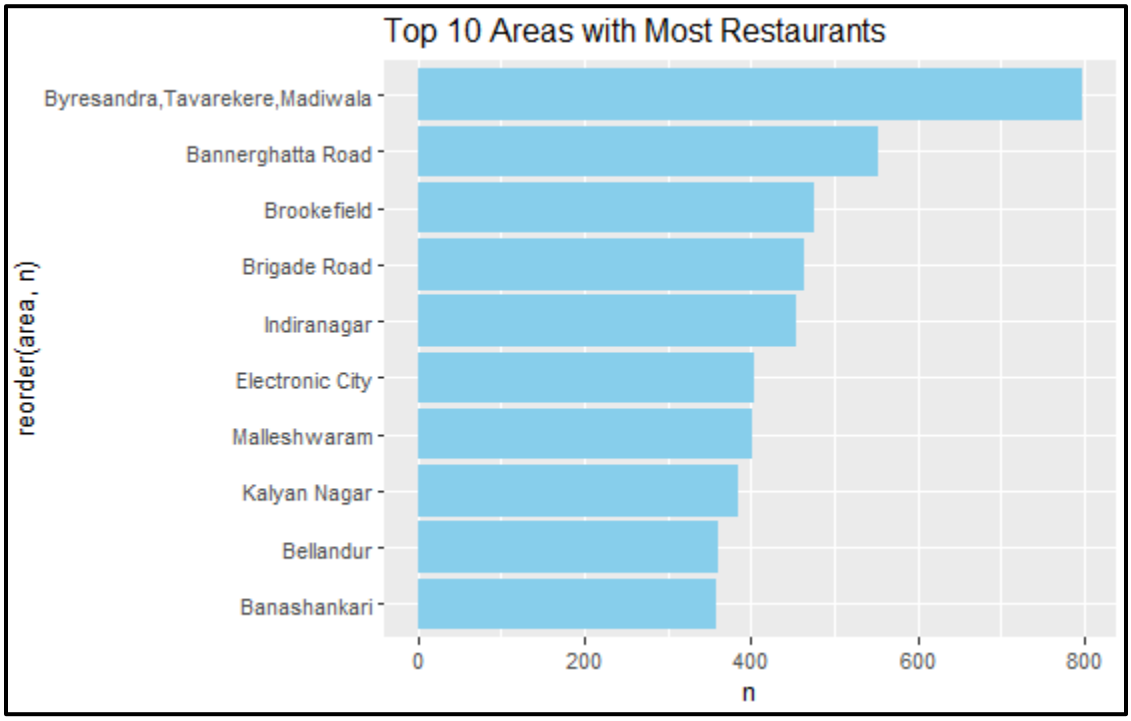
9. Analysis of Areas with the Highest Concentration of Restaurants

Analyzing restaurant density by area can reveal zones with high competition or strong demand for dining options. Restaurants considering entering these locations should take into account the competitive landscape and may consider differentiation strategies to stand out. Alternatively, targeting less saturated areas may capture unmet customer demand.

R Code:

```
ggplot(df %>% count(area, sort = TRUE) %>% head(10),
       aes(x = reorder(area, n), y = n)) +
  geom_bar(stat = "identity",fill="skyblue") + coord_flip() +
  labs(title = "Top 10 Areas with Most Restaurants")
```

Output:



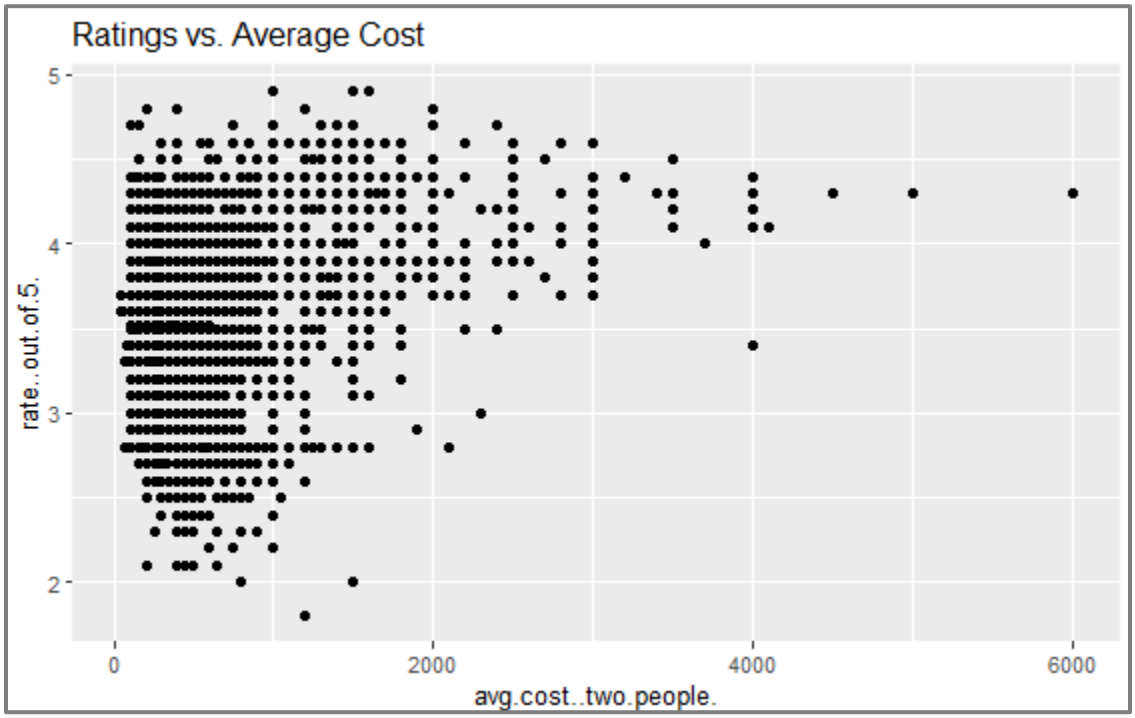
10. Examination of the Relationship Between Ratings and Average Cost

By examining the relationship between ratings and average cost, this analysis seeks to understand whether customers perceive higher-priced restaurants as better in quality. A weak correlation would suggest that factors other than pricing, such as service quality and ambiance, may be more influential in achieving high ratings.

R Code:

```
ggplot(df, aes(x = avg.cost..two.people., y = rate..out.of.5.)) +
  geom_point() +
  labs(title = "Ratings vs. Average Cost")
```

Output:



11. Data Analysis

A. Statistics for Average Cost

```
# Calculate specific descriptive statistics for a column
avg_cost_stats <- df %>%
  summarise(
    mean_cost = mean(avg.cost..two.people., na.rm = TRUE),
    median_cost = median(avg.cost..two.people., na.rm = TRUE),
    sd_cost = sd(avg.cost..two.people., na.rm = TRUE)
  )
print(avg_cost_stats)
```

```
> print(avg_cost_stats)
  mean_cost median_cost  sd_cost
1  540.2865         400 461.0415
```

These descriptive statistics provide a concise summary of the cost distribution across restaurants. Understanding the mean, median, and variability in dining costs allows stakeholders to assess whether they are positioned appropriately in terms of affordability or luxury. Additionally, it gives insight into market segmentation, helping businesses tailor their pricing strategy based on the market's current pricing trends.

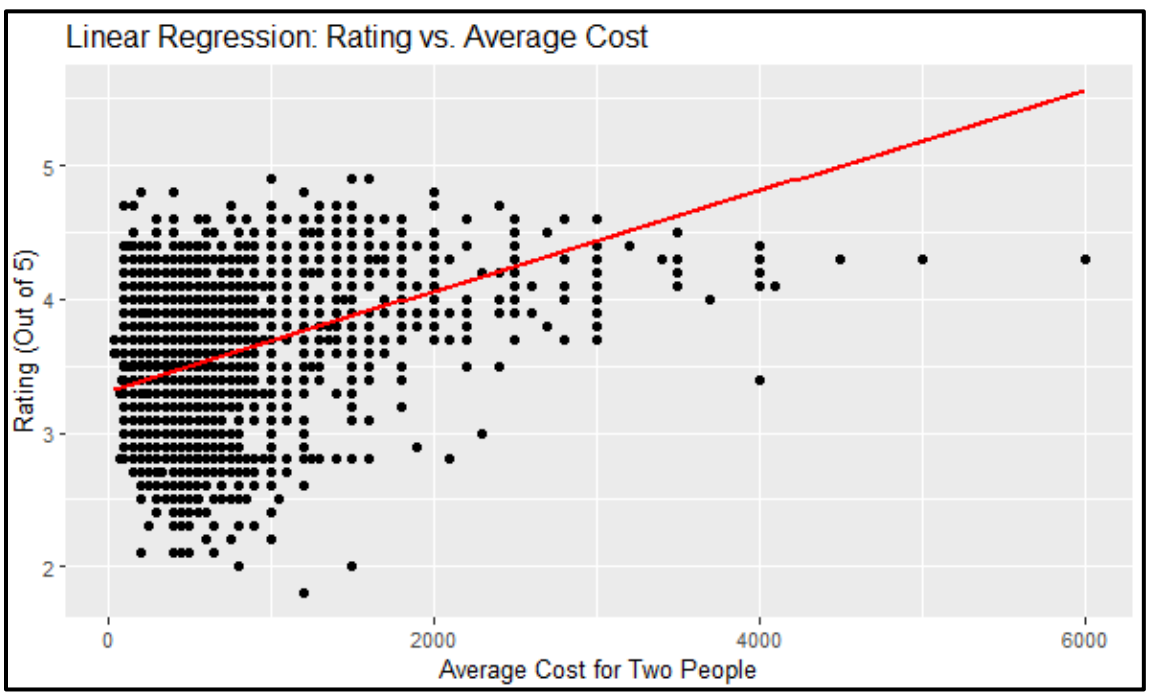
B. Linear Regression Model

This simple linear regression analysis explores the relationship between customer ratings (`rate..out.of.5.`) and the average cost for two people (`avg.cost..two.people.`). It aims to determine if pricing is a significant predictor of customer satisfaction.

```
# Simple linear regression model
linear_model <- lm(rate..out.of.5. ~ avg.cost..two.people., data = df)
summary(linear_model)
```

```
> summary(linear_model)

Call:
lm(formula = rate..out.of.5. ~ avg.cost..two.people., data = df)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.96103 -0.28696  0.03822  0.31304  1.41304 

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)    
(Intercept)           3.3121462  0.0078151   423.8   <2e-16 ***
avg.cost..two.people. 0.0003741  0.0000110    34.0   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.4276 on 7103 degrees of freedom
Multiple R-squared:  0.1399,	Adjusted R-squared:  0.1398 
F-statistic:  1156 on 1 and 7103 DF,  p-value: < 2.2e-16
```

```
# Visualize the linear regression line
ggplot(df, aes(x = avg.cost..two.people., y = rate..out.of.5.)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  labs(title = "Linear Regression: Rating vs. Average Cost", x = "Average Cost for Two People", y = "Rating (Out of 5)")
```



- **Findings**: There is a statistically significant positive relationship between price and ratings. Higher-priced restaurants tend to have slightly higher ratings, though the effect is modest.

- **Implications**: While pricing influences ratings, other factors like service and food quality are likely more impactful on customer satisfaction. This analysis suggests that pricing alone is not sufficient to drive higher ratings.

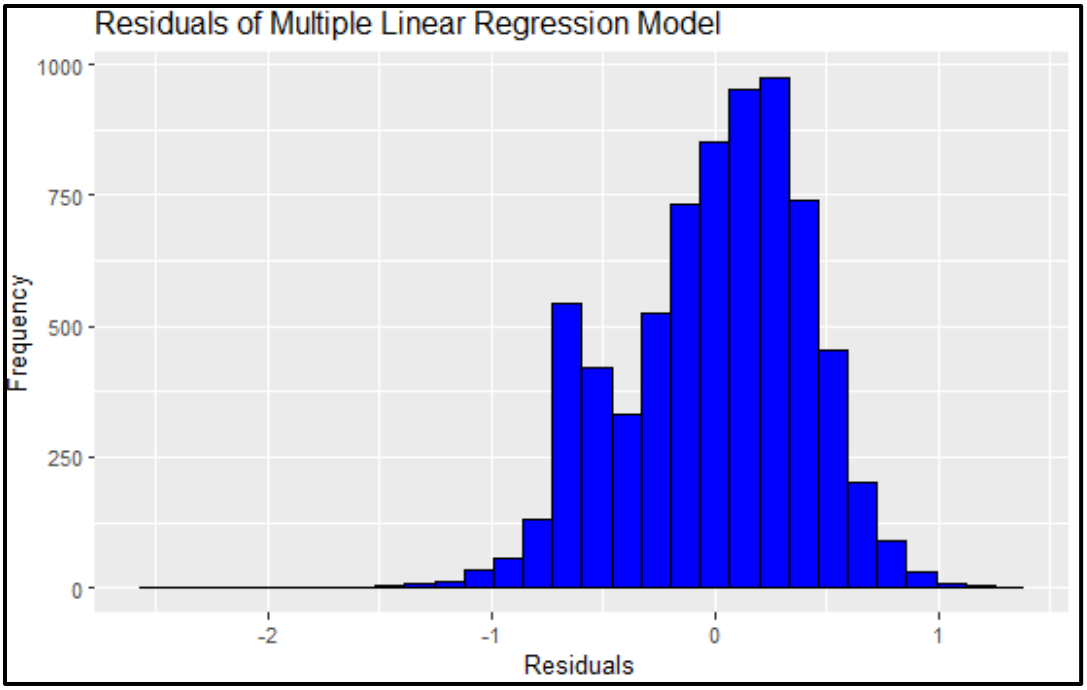
C. Multiple Regression Method

This multiple linear regression analysis examines the combined effect of average cost (`avg.cost..two.people.`) and the number of ratings (`num.of.ratings`) on customer ratings (`rate..out.of.5.`). This analysis seeks to determine if both price and popularity (measured by the number of ratings) influence customer satisfaction.

```
# Multiple linear regression model
multiple_linear_model <- lm(rate..out.of.5. ~ avg.cost..two.people. + num.of.ratings, data = df)
summary(multiple_linear_model)
```

```
> summary(multiple_linear_model)

Call:
lm(formula = rate..out.of.5. ~ avg.cost..two.people. + num.of.ratings,
    data = df)

Residuals:
     Min       1Q   Median       3Q      Max
-2.50488 -0.26237  0.05236  0.29121  1.32523

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)          3.323e+00  7.491e-03  443.55   <2e-16 ***
avg.cost..two.people. 2.768e-04  1.120e-05   24.72   <2e-16 ***
num.of.ratings        2.227e-04  8.719e-06   25.54   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.4092 on 7102 degrees of freedom
Multiple R-squared:  0.2123,    Adjusted R-squared:  0.2121
F-statistic: 956.9 on 2 and 7102 DF,  p-value: < 2.2e-16
```

```
# Plot residuals to check the fit
ggplot(data = data.frame(residuals = residuals(multiple_linear_model)), aes(x = residuals)) +
  geom_histogram(fill = "blue", color = "black", bins = 30) +
  labs(title = "Residuals of Multiple Linear Regression Model", x = "Residuals", y = "Frequency")
```



- **Findings**: Both average cost and number of ratings are statistically significant predictors of customer ratings. Higher costs and a greater number of ratings are associated with slightly higher customer ratings.
- **Implications**: This suggests that while higher costs and popularity (as reflected by the number of ratings) are linked to higher customer satisfaction, these factors together only account for about 21% of the variability. Other factors not included in this model, such as service quality, ambiance, and food type, are likely also important in determining customer satisfaction.

D. Predicting Customer Ratings for New Data

Objective:
Using the multiple linear regression model, we predict customer ratings based on hypothetical scenarios to understand how average cost and the number of ratings might influence predicted satisfaction.

```
# New data to predict ratings
new_data <- data.frame(avg.cost..two.people. = c(300, 500), num.of.ratings = c(100, 50))

# Predict ratings based on new data
predictions <- predict(multiple_linear_model, newdata = new_data)
print(predictions)
```

```
> predictions <- predict(multiple_linear_model, newdata = new_data)
> print(predictions)
       1        2
3.42794 3.47217
```

Interpretation:

- The predicted ratings provide insight into customer satisfaction under different cost and popularity conditions.
- A slight increase in rating is observed with a higher average cost, though it is not substantial, indicating that while cost and ratings influence satisfaction, their impact remains modest.

These predictions illustrate how the model can be used to forecast ratings for restaurants based on their average cost and number of reviews, supporting decision-making around pricing and customer engagement strategies

➢Conclusion

The analysis of the Zomato dataset provides valuable insights into customer preferences and satisfaction trends:

1. The analysis reveals that **"Quick Bites"** and **"Casual Dining"** dominate the restaurant market on Zomato. This suggests a high demand for informal, accessible dining experiences that appeal to a broad audience, making these restaurant types strategic choices for businesses looking to capture a large customer base.

2. This analysis highlights restaurants that cater to a premium clientele, revealing market positioning within the high-cost dining segment. These findings can guide strategies for attracting **high-income customers** by emphasizing luxury dining experiences.

3. The distribution indicates that while most restaurants receive a moderate number of ratings, a few have significantly high engagement. This suggests that these popular restaurants benefit from **strong customer loyalty**, which can be leveraged in future marketing and engagement strategies.

4. This analysis reveals that certain cuisines are particularly popular on Zomato, highlighting **strong customer demand**. Restaurants may benefit from featuring these cuisines to attract a broad customer base.

5. The findings identify restaurants that appeal to budget-conscious customers, providing an opportunity to market affordability to attract a wider audience.

6. The analysis reveals that the majority of restaurants offer online ordering, underscoring the **importance of digital convenience**. Adding online ordering options could increase a restaurant's appeal and competitiveness.

7. The findings show that table booking is more common in certain types of dining settings, suggesting that restaurants aiming to attract customers expecting a **formal dining experience** should offer this service.

8. This analysis highlights areas with high restaurant density, guiding new entrants to **evaluate competition** and **identify potentially lucrative locations**.

9. The scatter plot reveals a weak correlation between average cost and ratings, indicating that **higher pricing does not guarantee higher satisfaction**. This insight suggests that restaurants can enhance customer satisfaction through factors beyond price alone.

➢Recommendations

1. Focus on Popular Restaurant Types: Emphasizing "Quick Bites" and "Casual Dining" formats can capture a larger customer base due to their popularity.

2. Leverage Pricing Strategies: Target both premium and budget-conscious customer segments by tailoring pricing strategies.

3. Enhance Convenience: Embrace digital engagement by providing online ordering and table booking options to meet customer expectations.

4. Select Strategic Locations: High-density areas may benefit from service differentiation, while targeting less competitive areas could address unmet demand.